Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model — qualifier level (Lista 7)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Notation (Benigno 2015). Lower-case letters are log deviations from the steady state; bars denote the long run. Households: $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$, $u = C^{1-\tilde{\sigma}^{-1}}/(1-\tilde{\sigma}^{-1})$, $v = L^{1+\eta}/(1+\eta)$, $\rho \equiv -\ln\beta$. Resources: $y = s_c c + g$, $\sigma \equiv \tilde{\sigma} s_c$. Firms: demand $Y(j) = (P(j)/P)^{-\theta}(C+G)$, technology $Y(j) = AL(j)$; a fraction α of prices is preset at P^e. Calibration unless stated: $\alpha = 0.75$, $\sigma = 0.5$, $\eta = 0.2$, $\theta = 8$, $\rho = 0.01$ (a quarter), $p^e = \bar{p} = 0$, $g = \bar{g} = 0$, no taxes.

Block 1 — Foundations: AD, AS, natural and efficient output, taxes

(Benigno §§1–5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) The slope κ of the AS curve $p - p^e = \kappa(y - y_n)$ under the calibration is:

A. 6.600

B. 0.733

C. 0.233

D. 2.200

Answer: B

Step — derivation. Flexible firms price at $\tilde{p} = w - a + \mu$; labor supply gives $w - p = \eta l + \sigma^{-1}(y - g)$ with $l = y - a$; so $\tilde{p} - p = (\sigma^{-1} + \eta)(y - y_n)$. With $p = \alpha p^e + (1-\alpha)\tilde{p}$: $\kappa = \frac{(1-\alpha)(\sigma^{-1}+\eta)}{\alpha} = \frac{0.25 \cdot 2.2}{0.75} = 0.733$. A inverts $(1-\alpha)/\alpha$; C uses $\sigma + \eta$; D omits the rigidity factor.

Ref: Benigno (2015), §4, eqs. (16)–(17), pp. 506–508.

b) (3 points) *AS–AD × taxation.* Productivity rises by $a = 0.02$ and a payroll tax raises the (log) markup by $\mu = 0.05$, both temporarily. The changes in $(y_n,\ y_e)$ are:

A. $(-0.0227,\ 0)$

B. $(0.0109,\ 0.0109)$

C. $(-0.0118,\ -0.0118)$

D. $(-0.0118,\ 0.0109)$

Answer: D

Step — coefficients. $\sigma^{-1} + \eta = 2.2$. From (15): $\Delta y_n = (1.2 \cdot 0.02 - 0.05)/2.2 = -0.0118$; from (19): $\Delta y_e = 1.2 \cdot 0.02/2.2 = 0.0109$. Taxes enter only the markup (eq. 13), so they move y_n, never y_e.

Ref: Benigno (2015), §4.4, eqs. (13), (15), (19), pp. 507–509.

c) (2 points, T or F) Because θ sets the markup, a higher demand elasticity θ makes the AS curve steeper.

Answer: F

Step — why. θ enters μ, hence y_n, but not $\kappa = (1-\alpha)(\sigma^{-1} + \eta)/\alpha$: with linear technology and an economy-wide labor market there is no real rigidity. θ reappears in the welfare weight of the loss function (Question 4).

Ref: Benigno (2015), §4, eq. (17); Romer (2012), §6.5–6.7.

d) (2 points, T or F) *AS–AD × fiscal news.* Holding i, $\bar{p}$ and p^e fixed, news that long-run government spending will be higher ($\bar{g} \uparrow$, g unchanged) lowers current output.

Answer: T

Step — channel. In the AD, $\bar{y}_n - \bar{g}$ falls by $\frac{\eta}{\sigma^{-1}+\eta}\Delta\bar{g}$ (higher future taxes lower long-run consumption); consumption smoothing lowers current demand; the AS does not move, so y falls by $m_{\bar{g}}\Delta\bar{g}$, $m_{\bar{g}} = \eta/D = 0.067$, $D = (\sigma^{-1}+\eta)(1+\kappa\sigma)$.

Ref: Benigno (2015), §8, Table 1, p. 515.

Question 2. (15 points) Discursive. *AS–AD foundations × taxation × supply shocks.* Build the model from its primitives (no taxes unless stated).

a) (3 points) From the household problem with budget $PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$, derive the Euler equation and the labor-supply condition; log-linearize and combine with $y = s_c c + g$ to obtain the AD curve. Give its slope in the (y, p) plane, explain it, and list its shifters.

Step — FOCs. Lagrangian with multiplier λ: $u_c(C) = \lambda P$, $\beta u_c(\bar{C}) = \lambda\bar{P}/(1+i)$, $v_l(L) = \lambda W$; hence $\frac{u_c(C)}{\beta u_c(\bar{C})} = (1+i)\frac{P}{\bar{P}} \equiv 1 + r$ and $\frac{v_l(L)}{u_c(C)} = \frac{W}{P}$.

Step — log-linear Euler. $u_c = C^{-1/\tilde{\sigma}}$: $\bar{c} - c = \tilde{\sigma}[i - (\bar{p} - p) - \rho]$ (with $\ln(1+i) \approx i$, $\rho = -\ln\beta$).

Step — AD. $c = (y - g)/s_c$ and $\bar{c} = (\bar{y} - \bar{g})/s_c$, and long-run output is $\bar{y}_n$:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma[i - (\bar{p} - p) - \rho].$$

Slope $dp/dy = -1/\sigma$: a higher p lowers expected inflation $\bar{p} - p$, raises the real rate and current saving (an intertemporal channel, no LM curve). Shifters: i, g, $\bar{g}$, $\bar{p}$, $\bar{y}_n$ (i.e. $\bar{a}$, $\bar{\mu}$, $\bar{g}$).

Ref: Benigno (2015), §3, eqs. (1)–(6) and (21), pp. 505–509.

Grading rubric.

- FOCs: 1 pt.
- Log-linear Euler: 1 pt.
- AD, slope with its mechanism, shifters: 1 pt. Slope justified by money demand: 0 for this line.

b) (3 points) From the firm's problem derive the optimal flexible price, natural output and the AS curve with a fraction α of preset prices.

Step — optimal price. $\Pi(j) = (P(j) - \frac{W}{A})(\frac{P(j)}{P})^{-\theta}(C+G)$; FOC: $(1-\theta)P(j)^{-\theta} + \theta\frac{W}{A}P(j)^{-\theta-1} = 0 \Rightarrow P(j) = \frac{\theta}{\theta-1}\frac{W}{A}$; in logs $\tilde{p} = w - a + \mu$.

Step — gap relation. Labor supply in logs, $w - p = \eta(y-a) + \sigma^{-1}(y-g)$ (using $l = y - a$, $c = (y-g)/s_c$), gives $\tilde{p} - p = (\sigma^{-1}+\eta)y - (1+\eta)a - \sigma^{-1}g + \mu = (\sigma^{-1}+\eta)(y - y_n)$ with $y_n = \frac{(1+\eta)a + \sigma^{-1}g - \mu}{\sigma^{-1}+\eta}$ (where $\tilde{p} = p$).

Step — AS. $p = \alpha p^e + (1-\alpha)\tilde{p} \Rightarrow \alpha(p - p^e) = (1-\alpha)(\sigma^{-1}+\eta)(y - y_n)$: $\kappa = \frac{(1-\alpha)(\sigma^{-1}+\eta)}{\alpha} = 0.733$.

Ref: Benigno (2015), §4, eqs. (9)–(17), pp. 506–508; Romer (2012), problem 6.16.

Grading rubric.

- Optimal price: 1 pt.
- Gap relation and y_n: 1 pt.
- AS and κ: 1 pt. Quoting κ: 0 for this line.

c) (3 points) Derive efficient output from the planner's problem and compare it with natural output. Show how labor, payroll and consumption taxes affect each, and find the subsidy that makes the steady state efficient.

Step — planner. $\max u(Y-G) - v(Y/A)$: $u_c(Y-G) = v_l(Y/A)/A$; in logs $\sigma^{-1}(y-g) + \eta(y-a) = a$, so $y_e = \frac{(1+\eta)a + \sigma^{-1}g}{\sigma^{-1}+\eta}$ and $y_n - y_e = -\frac{\mu}{\sigma^{-1}+\eta}$: the markup is a wedge between the real wage and the marginal product of labor.

Step — taxes. With taxes the flexible price becomes $P(j) = \frac{\theta}{\theta-1}\frac{1+\tau_w}{1-\tau_y}\frac{W}{A}$ and the labor-supply condition carries $\frac{1-\tau_l}{1+\tau_c}$; both wedges add up in the aggregate markup $1 + \mu = \frac{\theta}{\theta-1}\frac{(1+\tau_w)(1+\tau_c)}{(1-\tau_y)(1-\tau_l)}$ (eq. 13). Taxes therefore lower y_n and leave y_e untouched: the planner faces no tax wedge.

Step — efficient subsidy. $1 + \mu = 1$ with only a revenue subsidy: $\tau_y = -\frac{1}{\theta-1}$, i.e. 14.3% at $\theta = 8$.

Ref: Benigno (2015), §4.4, eq. (19), p. 509; eq. (13), p. 507; footnote 21.

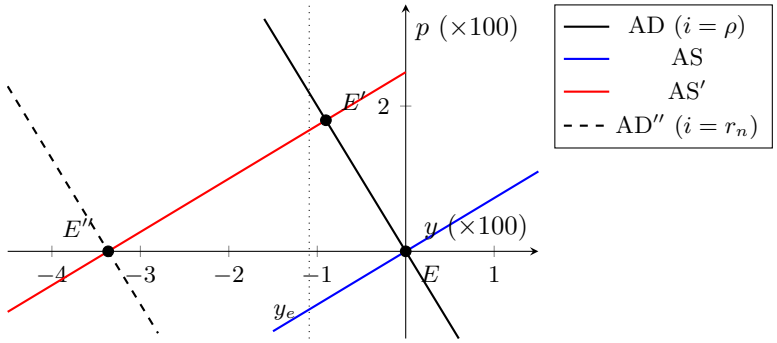
Grading rubric.

- Planner and y_e: 1 pt.
- Taxes in the markup, effect on y_n only: 1.5 pts.
- Subsidy: 0.5 pt.

d) (3 points) An oil shock raises the markup by $\mu = 0.05$ while productivity falls by $a = -0.02$, both for the short run only. Find the equilibrium output gap in terms of the natural real rate and the policy rate, compute the outcome if the bank keeps $i = \rho$ and the i that delivers $p = p^e$, and draw the AS–AD diagram (old and new AS, AD, both equilibria and y_e).

Step — gap. Substituting the AS into the AD ($g = \bar{g}$): $(1+\sigma\kappa)(y-y_n) = \bar{y}_n - y_n - \sigma(i-\rho) + \sigma(\bar{p}-p^e)$, i.e. $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n - i + (\bar{p} - p^e)]$ with $r_n \equiv \rho + \sigma^{-1}(\bar{y}_n - y_n)$.

Step — numbers. $y_n = -0.0336$, $y_e = -0.0109$, $r_n = 0.01 + 2(0.0336) = 7.73\%$. With $i = \rho$: gap $= 0.366 \cdot 0.0673 = 0.0246$, $y = -0.0090$, $p - p^e = 0.0180$; output is above efficient ($y - y_e = +0.0019$). With $i = r_n = 7.73\%$: $p = p^e$, $y = y_n$, 2.3% below y_e.



Ref: Benigno (2015), §5 and §7, pp. 509–514; natural rate on p. 513.

Grading rubric.

- Gap formula derived: 1 pt.
- Numbers for both policies: 1 pt.
- Graph: AS shifts through (y_n', p^e), AD unchanged at $i = \rho$, E' and E'' placed correctly, y_e marked: 1 pt.

e) (3 points) *Parameter analysis.* For the markup shock alone ($\mu = 0.05$, $a = 0$, $i = \rho$), how do the responses of output and prices depend on price rigidity? Find them in closed form as functions of κ, prove the signs, give the limits $\alpha \to 0$ and $\alpha \to 1$, fill the table for $\alpha \in \{0.5, 0.66, 0.75, 0.9\}$ and read it.

Step — closed forms. $y_n = -\frac{\mu}{\sigma^{-1}+\eta}$, $r_n - \rho = \frac{\sigma^{-1}\mu}{\sigma^{-1}+\eta}$, so the gap is $\frac{\mu}{(1+\sigma\kappa)(\sigma^{-1}+\eta)}$ and

$$y = -\frac{\mu}{\sigma^{-1}+\eta} \cdot \frac{\sigma\kappa}{1+\sigma\kappa}, \qquad p - p^e = \frac{\mu}{\sigma^{-1}+\eta} \cdot \frac{\kappa}{1+\sigma\kappa}.$$

Step — signs and limits. $\frac{\partial}{\partial\kappa}\frac{\kappa}{1+\sigma\kappa} = \frac{1}{(1+\sigma\kappa)^2} > 0$: more flexibility (lower α, higher κ) means a larger fall in output and a larger rise in prices. $\alpha \to 1$ ($\kappa \to 0$): $y \to 0$, $p \to p^e$ (output is demand-determined). $\alpha \to 0$ ($\kappa \to \infty$): $y \to y_n$ and $p - p^e \to \frac{\mu}{1+\sigma\eta} = 4.5\%$.

Step — table.

α	0.5	0.66	0.75	0.9
κ	2.200	1.133	0.733	0.244
y (%)	−1.19	−0.82	−0.61	−0.25
$p - p^e$ (%)	2.38	1.64	1.22	0.50

Step — reading. Rigid prices shield output from a cost shock when the policy rate is held fixed, because demand pins output down; flexible prices pass the shock to prices and output falls toward y_n. With the markup shock alone y_e does not move, so every response leaves output below the efficient level, and the more flexible the prices, the larger both the efficient gap and the price surprise (in (d), output could exceed y_e only because y_e fell too).

Ref: Benigno (2015), §7, p. 513; Romer (2012), problem 6.16.

Grading rubric.

- Closed forms: 1 pt.
- Signs and both limits proved: 1 pt.
- Table and reading: 1 pt.

Block 2 — Policy: optimal rules, the zero bound and household debt

(Benigno §§6–12 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Monetary × fiscal policy.* A temporary cut in consumption taxes of 2 points (like the 2009 IPI cut on cars, modeled as $\tau_c = -0.02$, entering the markup one for one) changes the natural real rate $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) + \sigma^{-1}(g - \bar{g}) + (\bar{\tau}_c - \tau_c)$ by:

A. +2.00 points

B. −1.82 points

C. +0.18 points

D. 0

Answer: C

Step — two channels. Supply: $\Delta\mu = -0.02$, $\Delta y_n = 0.02/2.2 = 0.0091$, lowering r_n by 0.0182. Demand: $\bar{\tau}_c - \tau_c$ rises by 0.02. Net +0.0018: the bank must raise i slightly to keep output at its natural level.

Ref: Benigno (2015), §8 and p. 513.

b) (3 points) *Fiscal multipliers.* With i fixed, a temporary rise in g of 1% of GDP changes output and the gap $y - y_n$ by, respectively:

A. 1.000% and 0.000%

B. 0.909% and 0.000%

C. 0.732% and 0.067%

D. 0.976% and 0.067%

Answer: D

Step — both curves move. $\Delta y_n = \frac{\sigma^{-1}}{\sigma^{-1}+\eta} = 0.909$ per unit; $\Delta r_n = \sigma^{-1}(1 - 0.909) = 0.182$; gap $= 0.366 \cdot 0.182 = 0.067$; output $0.909 + 0.067 = 0.976 = m_g$ (Table 1).

Ref: Benigno (2015), §8, Table 1, p. 515.

c) (2 points, T or F) At the zero lower bound with $r_n < 0$, a credible commitment to a long-run price level higher by $-r_n$ restores $y = y_n$ and $p = p^e$.

Answer: T

Step — gap formula. With $i = 0$, $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n + (\bar{p} - p^e)] = 0$ when $\bar{p} - p^e = -r_n$; then $p = p^e$.

Ref: Benigno (2015), §9, p. 517.

d) (2 points, T or F) *AS–AD × household debt.* In the borrower–saver economy with constant inflation between the short and the long run, more price flexibility reduces the output cost of a deleveraging shock.

Answer: F

Step — paradox of flexibility. The AD slopes up (Fisher debt deflation) and the output response $\frac{(1-\chi)\hat{d}}{\chi-(1-\chi)d_0\kappa}$ grows in size with κ (Question 4(e)).

Ref: Benigno (2015), §10, pp. 519–521.

Question 4. (15 points) Discursive. *Optimal policy × zero lower bound × household debt.*

a) (3 points) The bank minimizes Benigno's welfare loss $L = \frac{1}{2}(y-y_e)^2 + \frac{1}{2}\frac{\theta}{\kappa}(p-p^e)^2$ (eq. 33) subject to the AS. Find the optimal relation between the efficient gap and price surprises, describe it in the (y,p) plane and prove that shocks that leave μ unchanged create no trade-off.

Step — substitution. With $p-p^e = \kappa(y-y_n)$: $L = \frac{1}{2}(y-y_e)^2 + \frac{1}{2}\theta\kappa(y-y_n)^2$, strictly convex in y.

Step — targeting rule. FOC: $(y-y_e)+\theta\kappa(y-y_n)=0$, i.e. $(y-y_e)+\theta(p-p^e)=0$ (eq. 34); $y^* = \frac{y_e+\theta\kappa y_n}{1+\theta\kappa}$. In the plane: a line through (y_e, p^e) with slope $-1/\theta$; the optimum is its intersection with the AS.

Step — no trade-off. If μ is unchanged, $y_n = y_e$, and $y = y_e$, $p = p^e$ satisfies both the AS and the rule with $L = 0$ (divine coincidence); it is implemented by $i = r_n + (\bar{p} - p^e)$.

Ref: Benigno (2015), §11, eqs. (33)–(34), pp. 521–522.

Grading rubric.

- Substitution and FOC: 1.5 pts.
- Geometry: 0.5 pt.
- Divine coincidence with the argument: 1 pt.

b) (3 points) For the shock of Question 2(d), compute the optimal y, $p-p^e$ and i and compare with strict price stability. Then show that no Taylor rule $i = \bar{\imath} + \psi_y(y-y_e) + \psi_\pi(p-p^e)$ with a constant intercept implements the optimum under productivity shocks ($a = \pm 0.004$).

Step — optimum. $\theta\kappa = 5.867$: $y^* = \frac{-0.0109+5.867(-0.0336)}{6.867} = -0.0303$, $p-p^e = \kappa(y^*-y_n) = 0.0024$; check $(y^*-y_e)+\theta(p-p^e) = -0.0194+0.0194 = 0$.

Step — interest rate. $i = r_n - \frac{1+\sigma\kappa}{\sigma}(y^*-y_n) = 0.0773 - 2.733\cdot 0.0033 = 6.82\%$, between the passive 1% and strict price stability's 7.73%: the bank accepts a 0.24% price rise to cushion output.

Step — Taylor rule. Under a pure productivity shock $y_n = y_e$ and the optimum has zero gaps, so any Taylor rule prescribes $i = \bar{\imath}$. The optimum needs $i = r_n$: $a = 0.004$ gives $r_n = 0.56\%$, $a = -0.004$ gives 1.44%. One intercept cannot equal both: the rule must track the natural rate.

Ref: Benigno (2015), §11.

Grading rubric.

- Optimal y, p with the check: 1 pt.
- Optimal i and comparison: 1 pt.
- Taylor-rule argument: 1 pt.

c) (3 points) Pessimism: long-run productivity is expected to fall, $\bar{a} = -0.05$, with $a = 0$. Show that the zero bound binds, compute output and prices at $i = 0$ with $\bar{p} = p^e$, find the $\bar{p}$ that restores the optimum, give the limits as $\kappa \to \infty$ and $\kappa \to 0$, and discuss credibility.

Step — natural rate. $\bar{y}_n = \frac{1.2}{2.2}(-0.05) = -0.0273$, $y_n = y_e = 0$, $r_n = 0.01 + 2(-0.0273) = -4.45\% < 0$: $i = r_n$ is unavailable.

Step — at $i = 0$. Gap $= 0.366\cdot(-0.0445) = -0.0163$; $p-p^e = -0.0120$. Remedy: $\bar{p}-p^e = -r_n = 4.45\%$.

Step — limits and credibility. $\kappa \to \infty$: gap $\to 0$ and $p-p^e \to r_n = -4.45\%$; $\kappa \to 0$: $p = p^e$ and gap $= \sigma r_n = -0.0223$. Once the recession is over the bank prefers $\bar{p} = p^e$: the promise is time-inconsistent and works only if believed.

Ref: Benigno (2015), §6.3 and §9, pp. 513 and 517.

Grading rubric.

- r_n and the binding bound: 1 pt.
- Outcome at $i = 0$ and the required $\bar{p}$: 1 pt.
- Limits and credibility: 1 pt.

d) (3 points) With borrowers and savers (share χ of savers, initial debt d_0), Benigno's AD is

$$y = \bar{y}_n - \varphi[i - (\bar{p}_n - p) - \rho] + \tfrac{1}{\chi}(g - \bar{g}) + \tfrac{1-\chi}{\chi}[\hat{d} + d_0(p - p^e)], \qquad \varphi \equiv \frac{\sigma\chi + (1-\chi)d_0\beta}{\chi},$$

where $\hat{d}$ is the change in the debt limit. Find dy/dp when $\bar{p}_n$ is anchored and when inflation $\bar{p}_n - p$ is constant, name the two channels, and derive the spending multipliers in both regimes; compute them with $\chi = 2/3$, $d_0 = 1.2$, $\beta = 0.98$.

Step — slopes. Anchored: $dy/dp = -\varphi + \frac{(1-\chi)d_0}{\chi} = -\varpi$, $\varpi \equiv \sigma - \frac{d_0(1-\beta)(1-\chi)}{\chi} = 0.488$: the real-rate channel outweighs the Fisher channel (a higher p erodes nominal debt). Constant inflation: the real-rate term no longer depends on p, so $dy/dp = +\frac{(1-\chi)d_0}{\chi} > 0$: the AD slopes up.

Step — multipliers. Temporary g moves y_n by $\frac{g}{1+\sigma\eta}$ and the AD by g/χ. Anchored: $m = \frac{1/\chi + \varpi\kappa/(1+\sigma\eta)}{1+\varpi\kappa}$. Constant inflation: $m = \frac{1-(1-\chi)d_0\kappa/(1+\sigma\eta)}{\chi-(1-\chi)d_0\kappa}$, valid when $\chi > (1-\chi)d_0\kappa$.

Step — numbers. $\alpha = 0.75$: 1.344 and 1.964; $\alpha = 0.66$: 1.290 and 2.756 (Benigno's figures). Both exceed one: Ricardian equivalence fails for constrained borrowers.

Ref: Benigno (2015), §10, eq. (32), pp. 519–521.

Grading rubric.

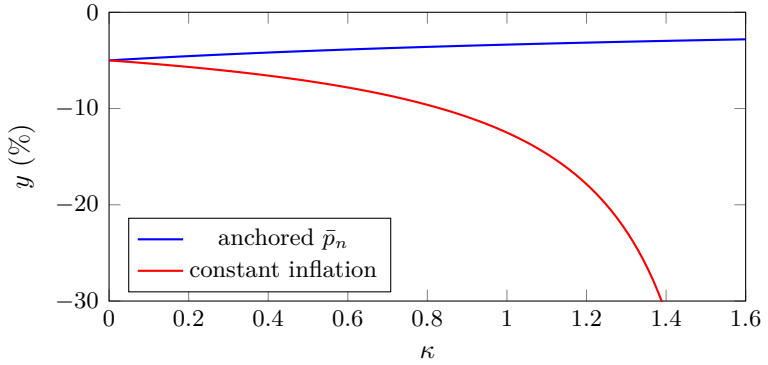
- Both slopes with the two channels: 1 pt.
- Both multipliers derived: 1.5 pts. Quoted from the article: max. 0.5 pt.
- Numbers: 0.5 pt.

e) (3 points) *Parameter analysis.* How does the output response to a fall in the debt limit $\hat{d} = -0.1$ depend on price flexibility in each regime? Find it in closed form, prove the signs, give the stability bound, fill the table for $\alpha \in \{0.66, 0.75, 0.9\}$, and draw both responses against κ.

Step — closed forms and signs. Anchored: $y = \frac{(1-\chi)\hat{d}/\chi}{1+\varpi\kappa}$, smaller in size as κ rises. Constant inflation: $y = \frac{(1-\chi)\hat{d}}{\chi-(1-\chi)d_0\kappa}$, larger in size as κ rises, and unbounded as $\kappa \to \frac{\chi}{(1-\chi)d_0} = 1.667$, the stability bound.

Step — table.

α (κ)	0.66 (1.133)	0.75 (0.733)	0.90 (0.244)
y anchored	−3.2%	−3.7%	−4.5%
y constant inflation	−15.6%	−8.9%	−5.9%
multiplier, anchored	1.29	1.34	1.44
multiplier, constant inflation	2.76	1.96	1.60



Step — reading. When long-run prices are anchored, flexible prices help: a lower p raises expected inflation and cuts the real rate. When inflation is expected to persist, a lower p lowers future prices too and raises the real value of debt: flexibility deepens the slump (paradox of flexibility), and the multiplier is largest exactly where the slump is.

Ref: Benigno (2015), §10, pp. 519–521, and footnote 20.

Grading rubric.

- Closed forms, signs and the bound: 1 pt.
- Table: 1 pt.
- Graph: one decreasing-in-size and one exploding curve, bound marked: 1 pt.